Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification of Architectural Choice

Problem Statement #28: Inter-City Freight Load Matching Marketplace

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Architecture Selection: *We chose Microservices Architecture for the Inter-City Freight Load Matching Marketplace System.*

Architectural Choice

Of the three styles evaluated in this lab, **Microservices** was selected over a Layered or Client-Server design. The Lab 1 use-case model already decomposes the marketplace into six loosely-coupled capabilities (Freight Load Posting, Carrier Bidding & Verification, Load Award & Contract Management, Shipment Tracking, Proof of Delivery & Escrow Settlement, and Platform Performance), and the Lab 2 backlog organised all thirteen user stories under exactly those six Epics. A microservices architecture lets each capability be built, deployed and scaled as its own service instead of being forced into shared horizontal layers or a single centralised server, which maps naturally onto how the requirements are already structured.

Two Reasons for This Choice

1. Independent scaling under peak auction load. NFR-002 requires the platform to support 5,000 concurrently open auctions with a 95th-percentile API response under 2 seconds. The Bidding & Verification Component carries this spike traffic and can be scaled out on its own, while lower-traffic services such as Shipment & POD tracking are left untouched, something a single layered application cannot do without scaling the entire system.
2. Fault isolation around two independent external integrations. The platform already depends on two external systems, the Carrier Verification Service and the Escrow Payment Gateway (per the Lab 1 actor list). Isolating each behind its own service means a slowdown or outage in the Escrow Payment Gateway only affects the Payment Service Component, while Load Posting, Bidding and Shipment Tracking continue operating normally.

Security Advantage

Each microservice is scoped to hold only the credentials and data it needs, following least-privilege. Only the Payment Service Component holds credentials for the Escrow Payment Gateway and only the Bidding & Verification Component talks to the Carrier Verification Service; a compromise of the public-facing Load Posting Component, for example, gives an attacker no direct path to escrowed funds or KYC data, which directly supports the encrypted, audit-logged escrow channel required by NFR-001.

Performance Benefit

Because the Bidding & Verification Component is deployed independently, it can be horizontally scaled precisely when auctions are closing (including UC-08 Auto-Extend Auction Window bursts) without redeploying or over-provisioning the rest of the system. This targeted scaling is what makes the NFR-002 target of p95 response under 2 seconds at 5,000 concurrent auctions realistically achievable.